

Challenges of Definition Extraction in Spanish Legal Texts

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Abstract—In this paper, we describe a preliminary experiment focused on definition extraction over Spanish legal texts based on a number of linguistic patterns and supported by an automatic term extraction tool. Through this experiment, we analyse current issues on the definition extraction task in general and, specifically, open challenges on the automatic identification of legal definitions in Spanish. As a result of this analysis, we also suggest various ideas on possible solutions to those issues and next steps in our work.

Index Terms—Definition Extraction, Legal Texts, Terminology

I. Introduction

Information Extraction (IE) and Information Retrieval (IR) are two areas of research in the Natural Language Processing (NLP) field whose aim is to obtain information from unstructured texts. Depending on the type of information to be identified, (be it names of companies or people mentioned in the text, terms dealt with in the document, or opinions given in Tweets), we will be referring to Named Entity Recognition, Term Extraction, or Opinion Mining, to mention but a few of the more specific tasks encompassed by these areas. When dealing with Term Extraction, the aim is to identify those terms or keywords that better define the content of a document. Terms may provide a network of superficial clues to understand whether a document fits the expected search criteria [5]. Automatic Term Acquisition may also be used for indexing new documents, or automatically annotating documents with the resulting terminology. In this sense, the richer the information describing the term, the more accurate the annotations will be.

With the purpose of adding definitions to terms extracted from a legal corpus, we have implemented a set of patterns to automatically discover definitions of Spanish legal terms implicitly defined in texts. It is well recognised that Legal language is a professional jargon, and, as such, it has some particularities not so frequently found in plain language. Spanish legal language is characterised by the frequent use

of technical terms, archaisms, and complex syntactic constructions (abundant periphrasis, long subordinate clauses, gerunds, impersonal and passive verbs, long enumerations, use of future subjunctive and future tense for obligation), preference for the 3rd person singular, aphoristic style, or the overuse of citations and references.

Definitions are also abundant in legal documents. Many terms will be directly defined in the text the first time they are mentioned. Some legal documents include a glossary of terms and definitions at the end of the document or as an annex (for example, European Directives and Regulations usually devote one article to define the main terms in the document¹). However, this is an uncommon practice in Spanish national law, and discovering definitions in text is a necessary and challenging task. The way in which legal terms are defined in legal discourse follows certain lexico-syntactic patterns that are not common in plain language.

In order to palliate the lack of approaches to automatically retrieve definitions of legal terms in Spanish legal documents, in this contribution we present an approach that strongly relies on lexico-syntactic patterns, understood here as a combination of lexical items following a certain syntactical structure that is recursively used to express a relation of interest, a definitional pattern in this case. Apart from the identification and implementation of those patterns, we have also worked in a previous automatic term extraction process over the corpus. In this regard, our objective was to test if this process improves the performance of the pattern extraction.

The rest of the document is structured as follows: firstly, we briefly refer to some works on the extraction of definitions in legal texts. Then, we refer to the main features of definitions in law, and describe the patterns we have created and their implementation. Finally, we discuss the results obtained with and without the help

¹<https://eur-lex.europa.eu/eli/reg/2016/679/oj>

of a terminology extraction tool, and refer to future lines of work.

II. Related work

The most commonly applied approaches to definition extraction are pattern based approaches. Their main drawback lays on the consuming process involved in the knowledge elicitation task; however, correctly built, patterns usually retrieve accurate results, as tested in various previous works. For instance, in 2004, Saggion applied a method that built the basis for a definition-based question-answering system [9]. The reproducibility of this work was, however, limited to a restricted number of definitions. In the case of [4], the authors propose patterns based on verbs and verbal phrases in Romanian, being the most productive ones those in which definitions were introduced by *denote*, *state*, *represent*, *define*, *specify*, *consist*, *name* and *permit*. A similar work, with similar results, is performed in [8] to retrieve definitions for Slavic languages. There have also been pattern-based approaches specialised in the legal domain, such as [10], in which the authors identify 52 rules based on lemma/POS and dependency parsing to extract definitions from a large corpus of German court decisions. To palliate the low recall they obtained, they used bootstrapping techniques based on single seed words at the expense of precision. The authors suggest that a balance between precision and recall is to be further achieved.

Leaving pattern- and rule-based approaches aside, there have also been several experiments relying on Machine Learning. The scalability of those works is always much higher than the one achieved with pattern-based approaches, but the manual work required for the annotation of a reference corpora is highly expensive. This is the case of [1], where they explored the use of machine learning to extract definitions from non-technical English texts, or [2], where results suggest that naive Bayes and random forest have the most suited learning bias for the task of definition extraction.

Mixed techniques have likewise been explored to identify definitions with higher linguistic variability than the classic encyclopedic *genus-et-differentia* definition. In 2009, Westerhout [11] proposed a definition extraction method for the automatic creation of glossaries within the area of eLearning. They used a sequential combination of a rule-based approach, based on POS-tags and a Machine Learning classification algorithm based on Balanced Random Forest. Another mixed work is [3], in which a weakly supervised bootstrapping approach was used to classify sentences in Wikidata as being definitions or not. In general, this kind of hybrid approaches show competitive results in

comparison with others only based on patterns, but there is no evidence of works in specialised languages or languages other than English.

All in all, most of the reviewed works suggest that identifying definitions is a challenging task that is dependant on the language and the area of expertise, and that probably mixed approaches are expected to provide effective solutions. Little work has been done with regard to legal definition extraction in Spanish. Our aim is to take the first steps in this topic through a preliminary experiment based on patterns and helped by an automatic terminology extraction tool. We analyse the main issues and challenges derived from this experiment, propose ideas to solve them and expose future steps.

III. Definitions in law

A *definition* is a piece of text that helps clarifying the specific sense of an expression that appears several times throughout a document [7]. Definitions are divided into two parts: *definiendum* and *definiens*. The *definiendum* is the word or element that is to be defined, and the *definiens* is the sentence or clause that gives the meaning to the *definiendum*. For example, in the definition "semantic is the science of meaning", the *definiendum* is "semantic" while the *definiens* is "the science of meaning". These parts are usually joined by a *copula*, which, in this case, would be "is". The most common structure of definitions is, hence, *definiendum*, *copula* and *definiens*.

However, these components are not always present in a fixed order. Specially in legal texts, it is common that the *definiens* component appears before the *definiendum*, and that the *copula* is not always expressed with the verb to be, which can sometimes lead to misunderstandings.

Specifically in Spanish legal texts, it is natural to find the structure *copula* + *definiendum* + *definiens*, for instance, "son documentos públicos los autorizados por un notario [...]" (*are public documents those authorised by a public notary**), where the *copula* is "son", the *definiendum* is "documentos públicos" and the *definiens* is "los autorizados por un notario". The *copula* can also be expressed with other verbs, commonly pronominal structures, as for instance "se considera" (*it is considered as*), "se entiende por" (*it is understood as*).

Notwithstanding, and depending on the legislator, other forms can also prevail in a text. For instance: "there is/exists" + *definiendum* + a condition expressed by "when", "if", "if and only if", or similar. In article 392 of the Spanish Civil Code, we can find the following definition: "hay comunidad cuando la propiedad

de una cosa pertenece a varias personas" (*there exists a community when the property of something belongs to several people*). And it is also not uncommon to find definitions introduced by some punctuation marks, that is, *Definiendum: Definiens*, or in apposition to the *Definiendum* between commas.

IV. Experiment

As theoretical foundation, we rely on the study of definitions in Spanish legal texts by [7]. Taking into account the different types of definitions that can be found in a legal text, we have identified the most standard patterns for legal definitions in Spanish (see Table I).

Our object of study is one of the most representative texts of Spanish labour law, the Spanish Workers' Statute². As a first step when applying linguistic patterns, we perform a part-of-speech tagging of the text. For that purpose, in this contribution we have used CoreNLP³ and spaCy⁴.

To identify the *definiendum*, we have followed two different strategies: 1) the one self generated by POS tokens and 2) the one assisted by an automatic term extraction tool. In the first case, we have generated sequences of lexico-syntactic patterns based on each token's part of speech (see Table II). In the second case, we have made use SketchEngine⁵ [6], a widely used corpora managing tool that also implements cutting-edge automatic term extraction algorithms. Our hypothesis is that by relying on the identification of the most relevant terms in the text, we may reduce the number of false positive results obtained, which identify as *definiendum* structures that do not correspond to any legal term.

After executing the two different approaches, we then compare the results obtained to check if previous terminology work can help in definition extraction, or if, on the contrary, identifying terms with part-of-speech tags works better in this specific case.

V. Results and Discussion

The main issue common to both approaches is, as expected, the great number of false positives retrieved. However, this study confirms the assumption that when relying on previously extracted terms, the number of false positives decreases. Still, in each approach we have been faced with different issues.

As for the definitions extracted with lexico-syntactic patterns, the major problem is caused by the erroneously assigned POS-tags. Lexico-syntactic patterns

rely completely on the accurate tagging of words in a text. Regarding those definitions retrieved after a previous term extraction task, we also rely on the accurate performance of the automatic term extraction tool, which may also wrongly identify tokens as terms. For instance, we retrieved the following definition: *A los anteriores efectos, se considerará salario la cantidad reconocida como tal en acto de conciliación o en resolución judicial por todos los conceptos a que se refiere el artículo 26* (For the above mentioned purposes, shall be considered salary the amount recognized as such in an act of conciliation or in a judicial resolution for all the concepts referred to in article 26*). The term defined in this case is "salario" (salary). However, the term extraction tool wrongly identified "salario la cantidad" (salary the amount) as the multi-word term being defined.

Similarly, we also retrieved: *El contrato de trabajo, se entenderá celebrado a tiempo parcial cuando se haya acordado la prestación de servicios durante un número de horas al día, a la semana, al mes o al año, inferior a la jornada de trabajo de un trabajador a tiempo completo comparable*. (*The employment contract shall be understood as part-time when the provision of services is agreed for a number of hours per day, per week, per month or per year, lower than the working day of a comparable full-time worker*). The *definiendum* in this case is "celebrado a tiempo parcial"; however, this term does not appear in our term list, and our patterns wrongly matched this definition with the term "tiempo parcial", that has a broader scope.. Therefore, the automatically extracted terms need to be post-processed, either by humans or with an automatic post-processing approach to avoid this kind of mistakes.

A different issue occurs when we find part of the term before the copula and part of the term after the copula. For instance, in this definition of the term fair dismissal, we have the term "despido" (dismissal) before the copula, and the modifying adjective "procedente" (fair) after the copula. See: *El despido se considerará procedente cuando quede acreditado el incumplimiento alegado por el empresario en su escrito de comunicación* (The dismissal will be considered fair when the breach alleged by the employer in his communication letter is proven*). A human can reason that the term defined is "despido procedente", but our algorithm cannot. Therefore, in this sentence, we have different term choices: *How can a machine reason which of the terms is being defined?* This is doubtlessly a very interesting challenge to work on.

To evaluate the performance of the patterns, with and without previous term extraction, we have created a gold standard by manually annotating the definitions

²<https://www.boe.es/eli/es/rdlg/2015/10/23/2/con>

³<https://stanfordnlp.github.io/CoreNLP/>

⁴<https://spacy.io/>

⁵<https://www.sketchengine.eu/>

TABLE I

Standard patterns with examples. *We have included literal translations in order to show the bilingual examples of the patterns.

	Spanish Patterns (and English translation)	Examples in English and Spanish
1	Hay/Existe + definiendum + cuando/siempre/if + definiens (There is/exists + definiendum + when/if + definiens)	<i>There is community when there are more than five people.*</i> <i>Existe una comunidad cuando hay más de 5 personas*</i>
1	Se considera + definiendum + definiens (It is considered + definiendum + definiens)	<i>It is considered collective agreement an agreement that...</i> <i>Se considera convenio colectivo un convenio que...</i>
3	Definiendum + es/son + definiens (Definiendum + is/are + definiens)	<i>Workers are people who work.</i> <i>Trabajadores son aquellas personas que trabajan.</i>
4	Es/son + definiendum + definiens (It/They is/are + definiendum + definiens)	<i>It is a worker the person who works.*</i> <i>Es un trabajador la persona que trabaja.*</i>
5	Se define + definiendum + como + definiens (It is defined + definiendum + as + definiens)	<i>It is defined worker as a person who works.*</i> <i>Se define trabajador como una persona que trabaja.*</i>
6	Se define + como + definiendum + definiens (It is defined + as + definiendum + definiens)	<i>It is defined as worker a person who works.*</i> <i>Es definido como trabajador una persona que trabaja.*</i>
7	Se entiende que + verb + definiendum + definiens (It is understood that + verb + definiendum + definiens)	<i>It is understood that there is a collective contract when...*</i> <i>Se entiende que hay contrato colectivo cuando...*</i>

TABLE II

Term patterns with approximate translations in English

	Term patterns	Term examples (with approximate English translation)
1	haber + noun	ley (law)
2	denominar + noun + conj + noun	empleador o empresario(employer or businessman)
3	considerarán + noun	familiares (familiar)
4	considerarán + noun + adj	relaciones laborales (labour relations)
5	considerará + noun + adp + noun	centro de trabajo (workplace)
6	considerará + adp + noun + adj	de carácter colectivo (collective character)
7	considerará + verb + det + noun	terminado el contrato (finished contract)
8	entender + adj + adv	prorroga automática (automatic extension)
9	entender + noun + adj	grupo profesional (professional group)
10	entender + noun + adp + noun	reducción de jornada (reduction of working hours)
11	entender + verb + adp + det + noun + adj	excluida de el ámbito laboral (excluded from the workplace)
12	entender + adj + adp + noun + adj	celebrado por tiempo indefinido (for indefinite period)
13	entender + adj + conj + adp + noun	nulos y sin efectos (null and void)
14	entender + verb + noun + adj	causas técnicas (technical reasons)
15	entender + verb + noun + adp + det + noun	insolvencia del empresario (insolvency of the employer)
16	entender + det + noun + aux + adj	la disminución es persistente (persistent decrease)
17	denominar + proppn + adp + proppn	código de trabajo (work code)
18	denominarán + noun	empresario (entrepreneur)
19	denominarán + noun + adj + adp + det + adj	personalidad jurídica de el contratante (legal entity of the contractor)
20	denominarán + noun + adp + det + noun	trabajadores de el contratista (workers of the contractor)
21	denominarán + noun + adp + noun	semanas de disfrute (weeks of leave)
22	denominarán + ponce + det + noun + conj	el contratista o subcontratista (contractor or subcontractor)

contained in the Spanish Workers' Statute. Results are exposed in table III The annotations were made by three linguist professionals, and they are available amongst the rest of the material involved in the experiment in our Github repository⁶.

We have also compared the automatically identified definitions with the ones provided in the most important source of legal lexicographic knowledge in Spain: Pan-Hispanic Dictionary of Legal Spanish (*Diccionario Panhispánico del Español Jurídico*, DPEJ henceforth)⁷. Additionally, we also note that the definitions of some terms in the dictionary slightly differ from the ones we

TABLE III

In this table we compare the number of definitions manually annotated in the gold standard and the number of definitions identified by the patterns, without and with a previous term extraction stage.

Gold Standard	Without term extraction	With term extraction
66 annotated definitions	25 true positives 11 false positives	34 true positives 3 false positives

provided in the Workers' Statute. In this regard, we assume that terms may be more specifically defined in the context of a certain legislation, and that we may find various definitions for the same term in different norms. We plan to further explore this issue. Examples of terms that are not included in the DPEJ are:

⁶https://github.com/krnlet/legal_definition_extraction

⁷<https://dpej.rae.es/>

- Trabajador nocturno (night worker): *Para la aplicación de lo dispuesto en el párrafo anterior, se considerará trabajador nocturno a aquel que realice normalmente en periodo nocturno una parte no inferior a tres horas de su jornada diaria de trabajo, así como a aquel que se prevea que puede realizar en tal periodo una parte no inferior a un tercio de su jornada de trabajo anual.* (For the application of the provisions in the previous paragraph, it is considered a night worker who normally performs at least three working hours per day or a third of their working hours per year during the night.)
- Familiar (family member or relative): *Se considerarán familiares, a estos efectos, siempre que convivan con el empresario, el cónyuge, los descendientes, ascendientes y demás parientes por consanguinidad o afinidad, hasta el segundo grado inclusive y, en su caso, por adopción.* (For these purposes, they will be considered family members, provided they live with the employer, the spouse, the descendants, ascendants and other relatives by consanguinity or affinity, up to and including the second degree and, where appropriate, by adoption.)
- Código de Trabajo (Employment Code): *El Gobierno, a propuesta del Ministerio de Empleo y Seguridad Social, recogerá en un texto único denominado Código de Trabajo, las distintas leyes orgánicas y ordinarias que, junto con la presente, regulan las materias laborales, ordenándolas en títulos separados, uno por ley, con numeración correlativa, respetando íntegramente su texto literal.* (The Government, at the proposal of the Ministry of Employment and Social Security, will collect in a single text called the Employment Code, the different organic and ordinary laws that, together with the present, regulate labour matters, ordering them in separate titles, one by law, with consecutive numbering, fully respecting its text.)

An additional advantage of performing definition extraction over legal texts is that we can identify when the same term has been defined differently in several legal documents, for comparison purposes, and also, when several (complementary) definitions are available for the same term in the same document, as is the case of "centro de trabajo", work centre, in the document at hand:

- 1) *A efectos de esta ley se considera centro de trabajo la unidad productiva con organización específica, que sea dada de alta, como tal, ante la autoridad laboral.* (For the purposes of this law, the productive unit with a specific organization,

which is registered, as such, before the labor authority is considered a work center).

- 2) *En la actividad de trabajo en el mar se considerará como centro de trabajo el buque, entendiéndose situado en la provincia donde radique su puerto de base.* (In the activity of work at sea, the ship will be considered as a work center, understood to be located in the province where its base port is located).

This kind of "contextual enrichment" seems highly valuable, since lexicographic resources such as the DPEJ usually contain only one and more generic definition.

VI. Conclusions and Future Work

In this experiment, we have tackled the case of standard definitions in Spanish legal texts. Our approach relies on POS-taggers that have shown imperfect for Spanish. Research on more accurate taggers and lemmatizers is therefore a requirement. Other issues have been thrown by our own lexico-syntactic patterns in combination with the terms previously identified, which sometimes generated wrong results and also need to be reviewed and adjusted for the domain at hand. With regard to the automatic term extraction tool applied, SketchEngine, we observed that several of the terms identified are not completely correct nor relevant in the text. Thus, another step to take is to test the performance of different automatic term extraction tools, which may apply statistical or linguistic approaches, and also to research on a post-processing algorithm to remove noise from those terms. A curated term list would surely improve the definitions extraction process.

On the other hand, in this experiment we have left aside the so-called *non-standard legal definitions*. Such definitions can be divided into four different groups: 1) indirect definitions, 2) conditional definitions, 3) incidental definitions and 4) meta-linguistic definitions. The first group includes regular definitions but expressed in a more indirect or verbose manner. The second group refers to definitions that have to comply with more than one condition: if P, then X is A only if X is B. Incidental definitions are those that have been exposed without intention, while exposing another statement. Finally, meta-linguistic definitions are those in which the *definiendum* can be both *definiendum* and *definiens*. Examples on these types of definitions can be found in [7].

It goes without saying that such definitions are much more difficult to extract than the standard ones. In fact, we have also tried to apply lexico-syntactic patterns for this purpose with little success (for the moment).

A clear future step is to work on the automatic identification of non-standard definitions in legal texts. A preliminary idea would involve relying on the manual annotation of non-standard definitions of the Spanish Workers' Statute to train a machine/deep learning engine to recognise them. However, we do not completely discard the use of linguistic patterns as well. A combination of both could also turn into an interesting and effective approach.

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